

Box 25.1 Indications for renal biopsy

- Proteinuria (significant proteinuria (>1 g/day or protein-to-creatinine ratio >100 mg/mmol)
- Acute nephritic syndrome (microscopic hematuria with any degree of proteinuria)
- Unexplained reduction in kidney function (native or transplant kidney)
- Renal manifestations of systemic disease

Box_25_1

Box 25.2 Features to be assessed by light microscopy in glomeruli

- Globally sclerotic glomeruli
- Size
- Cellularity: increased cells (hypercellularity) in mesangium (mesangial hypercellularity; NOTE: normal mesangial areas contain two to three cells), in capillary lumens (endocapillary hypercellularity), or in Bowman space (extracapillary hypercellularity or crescents)
- Capillary wall thickness and form (e.g., wrinkled, with double contours, spikes, spicules, craters, and holes; NOTE: best seen in periodic acid–Schiff or silver stain)
- Mesangial expansion (mesangial sclerosis) with/without nodules
- Deposition of abnormal material (e.g., amyloid)
- Segmental sclerosis
- Presence of thrombi, pseudothrombi, necrosis
- The distribution of the previously mentioned lesions (i.e., focal vs. diffuse and segmental vs. global)

Box_25_2

Box 25.3 Features to be assessed by light microscopy in tubules

- Percentage atrophy
- Signs of acute damage (e.g., dilatation, loss of brush border, epithelial flattening, mitoses, and necrosis)
- Tubulitis
- Casts: Granular casts suggest acute tubular injury; eosinophilic fractured casts suggest myeloma; neutrophil casts suggest acute pyelonephritis, pigmented cast (e.g., bile acids and myoglobin)
- Crystals (e.g., oxalate and calcium phosphate)
- Viral inclusions (e.g., BK virus)
- Distribution of the abovementioned lesions (i.e., focal vs. diffuse)

Box_25_3

Box 25.4 Examples of the use of electron microscopy (EM) in diagnosis of kidney biopsies

EM essential for diagnosis
Immunotactoid glomerulonephritis (GN)
Alport syndrome (including thin basement membrane lesion)
Fabry disease
Lecithin-cholesterol acetyltransferase deficiency
Nail patella syndrome
Fibrillary glomerulopathy (if DNAJB9 immunohistochemistry is not available)

EM helpful
Dense deposit disease
Minimal change disease
Early diabetic GN
Early membranous GN (particularly if only paraffin sections are available for immunohistochemistry)
Membranoproliferative GN
Postinfectious GN
Human immunodeficiency virus nephropathy
Lipoprotein GN
Collagenofibrotic GN

Diagnosis may be made without EM
IgA nephropathy
Acute tubulointerstitial nephritis
Myeloma cast nephropathy
Pauci-immune crescentic GN
Amyloid (although amyloid fibrils may be detected by EM when it has been missed on light microscopy)

Box_25_4

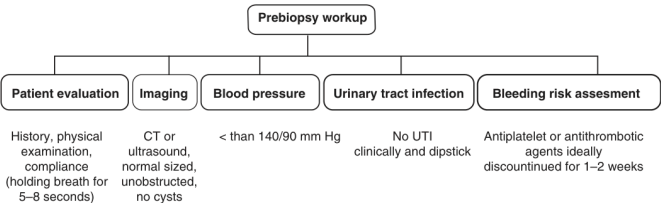


Fig. 25.1 Prebiopsy flowchart. CT, Computed tomography; UTI, urinary tract infection.

Fig_25_1

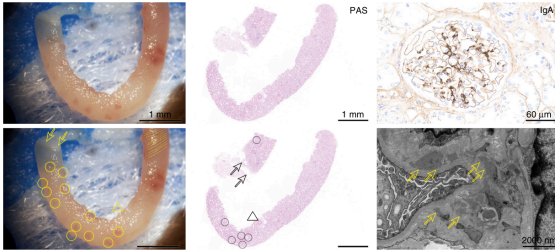


Fig. 25.2 Macroscopic appearance of kidney biopsy under a dissection microscope and its histopathologic workup on a case of IgA nephropathy. Under the stereomicroscope, the thick fibrous renal capsule (arrows, left image) is a good indicator of the cortical portion of the biopsy. It includes the glomeruli, which are visible as red spheres (circles, lower left image) in the fresh biopsy. The opposite part of the biopsy may contain intrarenal arteries (arrowhead) and medullary tissue indicated by red lines as a morphologic correlate of the vasa recta (dashed lines, bottom left image). For electron microscopy, a 1 × 1 × 1 mm piece of cortical tissue including glomeruli is removed and processed separately. The remaining tissue is dehydrated overnight. The dehydrated tissue is embedded in paraffin and cut into consecutive 1-µm thick cuts, mounted on a microscope slide. Slides are stained with various histochemical stainings including periodic acid–Schiff stain (PAS, middle image), highlighting glomerular basement membranes (circles, lower middle image), tubular basement membranes, and arterial walls (arrowhead). The subcapsular area (arrows, lower middle image) is separated from the biopsy core due to sampling for electron microscopy. Additionally, immunohistochemistry is applied to characterize immunoglobulin (e.g., mesangial IgA as shown by the dark brown color and upper right image) and complement deposition within glomeruli. Electron microscopy is used to detect electron-dense deposits (arrows, lower right image).

Fig_25_2

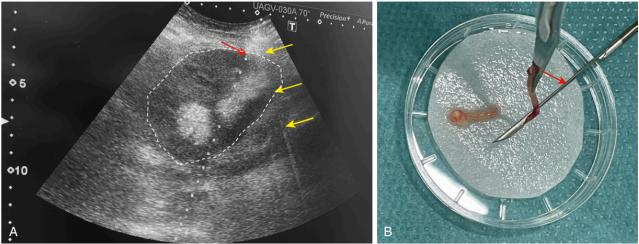


Fig. 25.3 Ultrasonography image during biopsy procedure and macroscopic appearance of biopsy. (A) Ultrasonography of a transplant kidney biopsy. The “flash” artifact from the biopsy needle is indicated by yellow arrows, while the tip of the biopsy needle is marked in red. (B) Macroscopic view of a kidney biopsy specimen retrieved from the biopsy needle (marked in red).

Fig_25_3

Table 25.1 Contraindications to renal biopsy

Absolute contraindications	Relative contraindications
Uncontrolled hypertension	Single kidney
Bleeding diathesis	Antiplatelet/clotting agents ^a
Widespread cystic disease	Anatomic abnormalities
Hydronephrosis	Small kidneys
Uncooperative patient	Active urinary/skin sepsis
	Obesity ^a

^aAspirin and body mass index >40 kg/m² were not found in recent limited cohort data to be a significant risk.

Table_25_1

Table 25.2 Alternative methods for obtaining renal tissue and their risks and benefits compared with a percutaneous approach

Method	Advantage	Disadvantage
Transjugular approach	Can be of use in those with a bleeding diathesis, ventilated patients, or if combined liver and renal biopsy is required	Risk of renal capsule perforation Inadequate material in up to 24%
Open approach	High yield of adequate tissue Hemostasis is more secure	Requires general/spinal anesthesia; longer recovery period
Laparoscopic approach	High yield of adequate tissue Hemostasis is more secure	Requires general/spinal anesthesia; longer recovery period

Table_25_2